L4b: Single Asset Geometric Brownian Motion Models

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Objectives for Today

Today we introduce our first continuous-time price model, geometric Brownian motion, estimate its parameters from data, test it against the stylized facts, and compute a closed-form target probability for a scheduled trade.

Today's concepts

- Model share prices with GBM: the stochastic differential equation, the Wiener process, the exact lognormal solution, and the exact one-step transition used for Monte Carlo.
- Estimate drift and volatility: derive the growth-rate law, estimate the mean growth rate and volatility from a growth-rate series, and recover the arithmetic drift.
- Test the model and evaluate a trade: check GBM against the stylized facts, then derive the probability that a scheduled long position clears a target return.

Lectures and Examples

Lecture:
CHEME-5660-L4b-Lecture-SingleAsset-GeometricBrownianMotion-TradeRule-Fall-2026.ipynb

Example 1:
CHEME-5660-L4b-Example-Parameters-SAGBM-Fall-2026.ipynb

Example 2:
CHEME-5660-L4b-Example-GBM-NPV-TradeRule-Fall-2026.ipynb

The first example estimates the mean growth rate and volatility of a GBM model from historical prices, recovers the drift, and simulates sample paths. The second computes the probability that a scheduled long position clears a target present-value return.

Company Profile: Jane Street

Jane Street is a privately held quantitative trading firm: it trades across asset classes on more than 200 venues, deploys mainly its own capital, and is a major market maker in exchange-traded funds.

- Client posture: Citadel Securities handles a large share of U.S. retail order flow; Jane Street is primarily proprietary and institution facing.
- Product emphasis: Jane Street is synonymous with ETFs; Citadel Securities with U.S. equities, options, and FX market making.
- Engineering culture: Jane Street optimizes for correctness and composability (functional programming in OCaml); Citadel Securities for reliability and throughput at scale.

Role in this lecture: both firms depend on quantitative models of prices and execution; today we build one baseline price model.

Concept Review: N-Ary Tree Models

A binomial lattice allows only two price movements per step. If two possible futures are not enough, an N-ary lattice allows m branches with factors $f_1, \dots, f_m$ and probabilities $p_1, \dots, p_m$.

- A recombining node is indexed by its branch-count vector; its probability is multinomial (L4a).
- More branches make the set of possible prices finer (at the cost of more states); a smaller time step brings the steps closer together.

Today's idea: replace the discrete set of prices with a continuous one and the discrete steps with continuous time. Refinement alone is not enough; with the right scaling the lattice converges to a continuous model, and that model is geometric Brownian motion.

Example: CHEME-5660-L4a-Example-N-Aray-Lattice-Fall-2026.ipynb

Single Asset Geometric Brownian Motion (GBM)

Let $S(t) > 0$ be the share price at time t on a horizon $0 \leq t \leq T$. Let $\mu \in \mathbb{R}$ (units: inverse years) be a constant arithmetic drift, $\sigma > 0$ (units: inverse years to the one-half power) a constant volatility, and $dW(t)$ the increment of a Wiener process (next frame). Geometric Brownian motion (GBM) is a continuous-time random walk whose drift and noise are both proportional to the price:

$$\frac{dS(t)}{S(t)} = \underbrace{\mu\, dt}_{\text{drift}} + \underbrace{\sigma\, dW(t)}_{\text{noise}}.$$

We will see that the log price is a Brownian motion with drift. We reserve g for an observed continuously compounded growth rate. The drift μ and a growth rate g share inverse-time units, but they are different quantities; the relationship is made precise below.

The Wiener Process

A Wiener process $\{W(t),\, 0 \le t \le T\}$ is a continuous real-valued stochastic process with three properties:

- The noise vanishes at zero: $W(0) = 0$ with probability one.
- Increments over disjoint intervals $W(t_1) - W(t_0), \dots, W(t_k) - W(t_{k-1})$ are independent for any $0 \le t_0 < t_1 < \cdots < t_k \le T$.
- Each increment is normal with mean zero and variance equal to the elapsed time, $W(t) - W(s) \sim \mathcal{N}(0, t-s)$ for $0 \le s < t \le T$.

The third property lets us write any increment as a scaled standard normal:

$$W(t) - W(s) = \sqrt{t-s}\, Z, \qquad Z \sim \mathcal{N}(0,1).$$

The Analytical Solution

Take $t_0 = 0$ and let S_0 be today's price. Solving the SDE with Itô calculus gives the share price at a fixed future time T:

$$\boxed{S_T = S_0 \exp\Big[\underbrace{\left(\mu - \tfrac{\sigma^2}{2}\right)T}_{\text{mean log growth}} + \underbrace{\sigma\sqrt{T}\,Z}_{\text{shock}}\Big], \qquad Z \sim \mathcal{N}(0,1).}$$

Thus the log price ratio $\ln(S_T/S_0)$ is normal with mean $(\mu - \sigma^2/2)T$ and variance $\sigma^2 T$, and the price S_T is lognormal and strictly positive.

The $-\sigma^2/2$ term comes out of the Itô calculus in the derivation notebook; it is the half-variance correction we meet again below.

Derivation: CHEME-5660-L4b-GBM-Solution-Derivation-Fall-2026.ipynb

Mean and Variance of the GBM Price

From the lognormal moments, the expectation and variance of S_T are:

$$\underbrace{\mathbb{E}(S_T)}_{\text{mean price}} = S_0 e^{\mu T}, \qquad \underbrace{\mathrm{Var}(S_T)}_{\text{price variance}} = S_0^2 e^{2\mu T}\left(e^{\sigma^2 T} - 1\right).$$

The expected price grows at the arithmetic drift μ. The median price $S_0 e^{(\mu-\sigma^2/2)T}$ grows at the smaller rate $\mu - \sigma^2/2$.

The gap is the half-variance correction: the lognormal distribution is right skewed, so its mean sits above its median.

Discrete Time GBM Model

We observe prices on a grid $t_j = j\Delta t$, $j = 0, 1, \dots, N$, with time step Δt (years) and horizon $T = N\Delta t$. Applying the fixed-horizon solution over one step gives the one-step transition:

$$\boxed{S_{t_j} = S_{t_{j-1}} \exp\Big[\big(\mu - \tfrac{\sigma^2}{2}\big)\Delta t + \sigma\sqrt{\Delta t}\, Z_j\Big], \qquad j = 1, \dots, N,}$$

with $Z_1, \dots, Z_N$ independent standard normals. For constant μ and σ this is exact at the grid points, not a numerical time-stepping approximation.

Jumping directly from S_0 to t_j uses the cumulative shock $\tilde{Z}_j = j^{-1/2} \sum_{i \le j} Z_i$:

$$S_{t_j} = S_0 \exp\Big[\big(\mu - \tfrac{\sigma^2}{2}\big)t_j + \sigma\sqrt{t_j}\, \tilde{Z}_j\Big].$$

$\tilde{Z}_j$ is standard normal at each j but not independent across j (correlation $\sqrt{t_j/t_k}$ for $j < k$): use it for one horizon, use the transition for paths.

Monte Carlo Simulation of GBM

Given S_0, μ, σ, Δt, N steps, and M paths, store the simulated prices in $\mathbf{S} \in \mathbb{R}^{(N+1)\times M}$, one path per column, rows indexed by $j = 0, \dots, N$.

For each sample path (each column of $\mathbf{S}$):

1. Set the price in row $j = 0$ to S_0.
2. For $j = 1, \dots, N$: draw $Z_j \sim \mathcal{N}(0, 1)$ and apply the one-step transition:

$$S_{t_j} \leftarrow S_{t_{j-1}} \exp\Big[\big(\mu - \tfrac{\sigma^2}{2}\big)\Delta t + \sigma\sqrt{\Delta t}\, Z_j\Big].$$

The result is M possible future trajectories of N steps each. Their spread is process uncertainty; the uncertainty in μ and σ is a separate layer.

Implementation: the `sample(...)` function in the course package.

Estimation of GBM Parameters

Divide the one-step transition by $S_{t_{j-1}}$, take logarithms, and divide by Δt. The one-step growth rate g_j over $t_{j-1} \to t_j$, the same quantity we computed from data in L3a, is:

$$\boxed{g_j \equiv \frac{1}{\Delta t} \ln \frac{S_{t_j}}{S_{t_{j-1}}} = \underbrace{\left(\mu - \frac{\sigma^2}{2}\right)}_{\text{mean growth } \bar{g}} + \underbrace{\frac{\sigma}{\sqrt{\Delta t}}}_{\text{growth-rate std}} Z_j.}$$

Because the Z_j are independent standard normals, growth rates over equally spaced, non-overlapping steps are independent and normal with mean $\bar{g} \equiv \mu - \sigma^2/2$ and variance $\sigma^2/\Delta t$.

- Drift versus growth: μ is the SDE drift; the mean growth rate $\bar{g}$ is what a growth-rate series measures. They differ by the half-variance correction.
- Risk: σ sets the growth-rate standard deviation $\sigma/\sqrt{\Delta t}$ and the one-step log-return standard deviation $\sigma\sqrt{\Delta t}$.

Volatility

From $N+1$ prices on the grid we form N growth rates $g_1, \ldots, g_N$. Their sample mean g' and sample standard deviation σ_g (our L3a risk measure) are:

$$g' = \frac{1}{N}\sum_{j=1}^{N} g_j, \qquad \underbrace{\sigma_g}_{\text{growth-rate std}} = \sqrt{\frac{1}{N-1}\sum_{j=1}^{N}(g_j - g')^2}.$$

(Denominator $N-1$: N observations less the one mean estimated from them.) But σ_g is not the GBM volatility. Since $\mathrm{Var}(g_j) = \sigma^2/\Delta t$, σ_g estimates $\sigma/\sqrt{\Delta t}$, and the volatility estimate is:

$$\boxed{\hat{\sigma} = \sigma_g\sqrt{\Delta t}.}$$

For daily data $\Delta t = 1/252$: divide the daily growth-rate standard deviation by $\sqrt{252}$, or equivalently multiply the daily log-return standard deviation by $\sqrt{252}$.

Drift

Under GBM the arithmetic drift is $\mu = \bar{g} + \sigma^2/2$. Given an estimate $\hat{\bar{g}}$ of the mean growth rate, the drift estimate is:

$$\underbrace{\hat{\mu}}_{\text{SDE drift}} = \underbrace{\hat{\bar{g}}}_{\text{mean growth}} + \underbrace{\tfrac{1}{2}\hat{\sigma}^2}_{\text{correction}} .$$

Two common ways to estimate $\bar{g}$:

- Method 1, average growth rate: $\hat{\bar{g}} = g'$, the sample mean, which is also the maximum likelihood estimate under GBM.
- Method 2, linear regression: regress log price on time; the slope estimates $\bar{g}$, but the residuals are a Brownian path, so the ordinary standard errors and residual checks do not apply.

The drift is the harder estimate: mean growth is small relative to growth-rate noise, so both methods carry substantial sampling error.

Linear Regression: The Model

Take the logarithm of the cumulative GBM solution at grid point t_j:

$$\ln S_{t_j} = \underbrace{\ln S_0}_{\text{intercept}} + \underbrace{\bar{g}\, t_j}_{\text{trend}} + \underbrace{\sigma W(t_j)}_{\text{error}}, \qquad j = 0, \dots, N.$$

The noise has mean zero, so the expected log price is a line with slope $\bar{g}$: regressing log price on time recovers that line, and the Wiener term is the error term.

That error is a Brownian path: neighboring values are strongly correlated, its variance $\sigma^2 t_j$ grows along the sample, and the fitted residuals inherit both features. Least squares still gives a sensible slope, but textbook standard errors, confidence intervals, and residual-normality tests assume independent, identically distributed errors and are not calibrated here.

Example 1: the Anderson–Darling normality test on these residuals rejects: a warning about the test's assumptions, not a verdict on GBM.

Linear Regression: The Normal Equations

Stack the $N+1$ log prices as an overdetermined system $\hat{\mathbf{X}}\theta + \epsilon = \mathbf{y}$, where θ holds the intercept $\ln S_0$ (left free, not pinned to day one) and the slope $\bar{g}$, ϵ is the error vector, and the augmented design matrix $\hat{\mathbf{X}}$ carries a column of ones:

$$\hat{\mathbf{X}} = \begin{bmatrix} 1 & t_0 \\ 1 & t_1 \\ \vdots & \vdots \\ 1 & t_N \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} \ln S_{t_0} \\ \ln S_{t_1} \\ \vdots \\ \ln S_{t_N} \end{bmatrix}.$$

The least-squares estimate is the solution of the normal equations:

$$\boxed{\widehat{\theta} = (\hat{\mathbf{X}}^{\top}\hat{\mathbf{X}})^{-1}\hat{\mathbf{X}}^{\top}\mathbf{y}.}$$

Its second component is the regression estimate $\widehat{\bar{g}}$.

Example 1: CHEME-5660-L4b-Example-Parameters-SAGBM-Fall-2026.ipynb

Stylized Facts

Does GBM capture the empirical regularities of asset growth rates?

- Heavy tails: empirical growth rates have heavier tails than a normal distribution, so extreme moves are more likely than a Gaussian predicts.
- Absence of autocorrelation: if prices follow a random walk, growth rates over non-overlapping intervals are uncorrelated, and there is no predictability to exploit.
- Volatility clustering: large moves tend to follow large moves and small moves follow small moves, so volatility alternates between high and low regimes.

We test each against the growth-rate law we just derived.

Growth Over a Horizon Under GBM

Define the horizon-average growth rate over $0 \to T$ as $g_{T,0} \equiv (1/T)\ln(S_T/S_0)$, in the same to-from subscript order as the discount factor $\mathcal{D}_{T,0}$. Substituting the GBM solution and dividing by T gives:

$$\boxed{\underbrace{g_{T,0}}_{\text{horizon-average growth}} = \bar{g} + \frac{\sigma}{\sqrt{T}}\, Z, \qquad Z \sim \mathcal{N}(0,1).}$$

So $g_{T,0}$ is normal with mean $\bar{g}$ and variance σ^2/T; the one-step case is $T = \Delta t$.

The uncertainty of the horizon-average growth rate shrinks as T grows while its mean stays put. The price does not settle down: $\ln(S_T/S_0) = T\, g_{T,0}$ has variance $\sigma^2 T$, which grows with T.

Heavy Tails and Volatility Clustering

Heavy tails: not captured. One-step and horizon-average growth rates are both normal under GBM. The lognormal price has a long right tail, but that is not a heavy-tailed growth-rate distribution.

Volatility clustering: not captured. The volatility σ is constant, so there is no mechanism for high- and low-volatility periods to alternate.

Both failures follow from the same assumption: independent Gaussian increments with a constant variance rate. That assumption is what makes the model exactly solvable.

Autocorrelation of One-Step Growth Rates

From the growth-rate law, $g_j = \bar{g} + (\sigma/\sqrt{\Delta t})Z_j$ with independent shocks. The autocovariance at lag $\tau \geq 1$ is:

$$\mathrm{Cov}(g_j, g_{j+\tau}) = \frac{\sigma^2}{\Delta t}\,\mathrm{Cov}(Z_j, Z_{j+\tau}) = 0,$$

because Z_j and $Z_{j+\tau}$ are independent. With $\mathrm{Var}(g_j) = \sigma^2/\Delta t$, the autocorrelation function is:

$$\boxed{\rho_g(\tau) = \frac{\mathrm{Cov}(g_j, g_{j+\tau})}{\mathrm{Var}(g_j)} = 0, \qquad \tau \geq 1.}$$

Absence of autocorrelation: captured for one-step growth rates over non-overlapping intervals. This says nothing about squared growth rates, whose dependence (L3a) GBM also cannot produce.

GBM Trade Rule

Buy $n_0 > 0$ shares at S_0 at time 0; sell all n_0 shares at S_T at the scheduled exit $T = N\Delta t$, $N \geq 1$ (frictionless: no costs, price return only). With annual continuously compounded benchmark growth rate g_b, $\mathcal{D}_{T,0}(g_b) = e^{g_b T}$ is the accumulation factor and $\mathcal{D}_{T,0}^{-1}(g_b) = e^{-g_b T}$ discounts the sale to time 0:

$$\mathrm{NPV}(g_b, T) = \underbrace{-\,n_0 S_0}_{\text{entry (now)}} + \underbrace{n_0 S_T\, \mathcal{D}_{T,0}^{-1}(g_b)}_{\text{exit (future)}}.$$

Dividing by the initial investment gives the scaled NPV, a dimensionless present-value return on the share cost:

$$\underbrace{\rho_T}_{\text{scaled NPV}} := \frac{\mathrm{NPV}(g_b, T)}{n_0 S_0} = \left(\frac{S_T}{S_0}\right) e^{-g_b T} - 1.$$

This is the L4a rule with one change: S_T now follows GBM instead of a binomial lattice.

Substitute the GBM Solution

Replacing S_T by the GBM solution puts all of the randomness in one standard normal Z:

$$\boxed{\rho_T = \exp\Big[(\mu - \tfrac{\sigma^2}{2})T + \sigma\sqrt{T}\,Z\Big]\,\underbrace{e^{-g_b T}}_{\text{discount}} - 1, \qquad Z \sim \mathcal{N}(0,1).}$$

- Short holding period: for a few days or weeks and a realistic benchmark, $|g_b|T$ is small, $e^{-g_b T} \approx 1$, and $\rho_T \approx S_T/S_0 - 1$ is the fractional price change.
- Longer holding period: whenever $|g_b|T$ is not small, keep the discount factor.

Either way ρ_T is a continuous random variable driven by Z, so a target probability is a one-line standardization.

Cumulative Probability of a Trade Exit

Let $\rho_\star > -1$ be a target (under GBM $S_T > 0$, so $\rho_T > -1$ always). We ask for the probability that ρ_T at the scheduled exit strictly exceeds the target. Moving the non-random terms to the right:

$$\mathbb{P}(\rho_T > \rho_\star) = \mathbb{P}\Big(\ln \tfrac{S_T}{S_0} > \ln(1+\rho_\star) + g_b T\Big).$$

Under GBM, $\ln(S_T/S_0)$ is normal with mean $(\mu - \sigma^2/2)T$ and standard deviation $\sigma\sqrt{T}$. Standardizing both sides:

$$\mathbb{P}(\rho_T > \rho_\star) = \mathbb{P}\left(\underbrace{\frac{\ln(S_T/S_0) - (\mu - \sigma^2/2)T}{\sigma\sqrt{T}}}_{Z\sim\mathcal{N}(0,1)} > \frac{\ln(1+\rho_\star) + g_b T - (\mu - \sigma^2/2)T}{\sigma\sqrt{T}}\right).$$

As in L4a, this is the scheduled terminal exit; a monitored take-profit or stop-loss is a first-passage question.

The Closed-Form Target Probability

A standard normal exceeds z with probability $1-\Phi(z)$, where Φ is the standard normal cumulative distribution function. Therefore:

$$\boxed{\mathbb{P}(\rho_T > \rho_\star) = 1 - \Phi\left(\frac{\ln(1+\rho_\star) + g_b T - (\mu - \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}\right).}$$

- The probability uses the real-world μ and σ estimated from data. A risk-neutral drift, which we will use to price derivatives later, changes the measure and no longer answers a probability-of-profit question.
- For a short holding period $g_b T \approx 0$ and the benchmark drops out of the numerator.
- A target of -1 or below is cleared with probability one.

Example 2: CHEME-5660-L4b-Example-GBM-NPV-TradeRule-Fall-2026.ipynb

Optional Advanced Material

Advanced 1: advanced/lattice-limit/
CHEME-5660-L4b-Advanced-LatticeToGBM-Fall-2026.ipynb

Advanced 2: advanced/first-passage/
CHEME-5660-L4b-Advanced-FirstPassage-GBM-Fall-2026.ipynb

Advanced 3: advanced/drift-uncertainty/
CHEME-5660-L4b-Advanced-DriftUncertainty-Fall-2026.ipynb

Advanced 4: advanced/monte-carlo/
CHEME-5660-L4b-Advanced-MonteCarlo-TargetProbability-Fall-2026.ipynb

Optional, not prerequisites for L5a: the L4a lattice calibrated to GBM and converging to the lognormal; first-passage take-profit and stop-loss rules under GBM; drift uncertainty and the span rule; Monte Carlo against the closed form.

Notebooks: lectures/week-4/L4b/advanced, also linked from the lecture notebook.

Summary

Today we introduced geometric Brownian motion as our first continuous-time price model, estimated its parameters from growth-rate data, tested it against the stylized facts, and derived a closed-form probability for a scheduled trade.

Key takeaways

- GBM is a tractable lognormal model: exact solution, exact one-step simulation with independent normal shocks, mean growing at the drift.
- Drift and growth are different parameters: the growth-rate series gives $\bar{g}$ and, through $\sigma_g\sqrt{\Delta t}$, the volatility; the drift follows from the half-variance correction.
- One stylized fact passed, two failed: growth rates are uncorrelated but Gaussian with constant volatility; the terminal target probability is one minus a standard normal CDF.

Next, we extend GBM to several correlated assets and use it to build portfolios.

Today: the lattice becomes a continuous process, and the trade rule gets a closed form.